

Mean-Force Hamiltonians from Influence Functionals

Gerard McCaul

(Dated: February 13, 2026)

The Hamiltonian of mean force (HMF) provides the standard starting point for strong-coupling thermodynamics, yet explicit operator forms are known only in restricted settings. We present a “quenched density” framework that uses the Hubbard-Stratonovich transformation to rewrite the reduced equilibrium state as an average over local propagators in imaginary time. This approach rigorously separates the statistical definition of the environment from the algebraic structure of the system response. We apply this framework to the minimal case of a harmonic environment with a coupling commuting with the system Hamiltonian. In this scenario the correction to the HMF has an exact, closed-form expression. We validate this result against finite-bath trace-out calculations and stochastic imaginary-time sampling in a five-level projector-coupled model. Finally, we outline the program to extend this framework to non-Gaussian and non-commuting environments.

I. INTRODUCTION

What governs equilibrium? Historically, the canonical answer is the thermal Gibbs state, where the environment is characterised by a single parameter: the temperature. Yet, this canonical answer is puzzling. It implies that neither the specific character of the environment nor the form of its coupling to the system alters the system’s equilibrium. This of course is a consequence of the assumptions used to derive the answer - the weak-coupling or Markovian regimes open-systems theory has historically relied upon. At strong coupling however, the reduced state inherits explicit temperature dependence and interaction-induced operator content that definitionally cannot be captured by these assumptions [1, 2]. These issues have motivated a broad investigation into strong-coupling thermodynamics, including exact fluctuation relations [3, 4], consistent heat and work definitions [5, 6], and operational measurability [7, 8].

Ultimately, while any subsystem in global equilibrium can formally be expressed in a canonical form, the effective Hamiltonian governing this state is generally not the bare Hamiltonian of the system. Instead, it must account for the non-trivial influence of the environment. If such an operator can be constructed, the standard description of equilibrium holds generically across all coupling strengths. *The Hamiltonian of mean force* (HMF) is the operator that reproduces this reduced equilibrium object as a Gibbs-like state, and provides the standard starting point for strong-coupling thermodynamics and mean-force Gibbs-state constructions [3, 9–11].

The ultimate target of thermodynamic descriptions is therefore unambiguous, and in fact a definition of HMF is equally unambiguous - it is simply the logarithm of the total state with the environment traced out. It is the *representation* of this operator that remains a challenge. Is the HMF a local operator? Does it admit a compact description in terms of few-body interactions? To address this, a veritable zoo of approximations - polaron transforms, weak-coupling expansions [12], reaction-coordinate mappings [13, 14] - have

proliferated strong-coupling thermodynamics. These approximations often disagree in intermediate regimes however, and while powerful numerical techniques such as hierarchical equations of motion (HEOM) or path-integral Monte Carlo exist [15–19], they typically do not yield compact operator forms for the HMF.

Ultimately, the source of these difficulties lie in the trace operation, which mixes the bath and system degrees of freedom in a way that is difficult to invert. For this reason, a general framework for representing the HMF has proved elusive. The purpose of this paper is to lay the groundwork for just such a framework. To do so, we construct a representation of the equilibrium density matrix that disentangles the statistical properties of the environment from the algebraic response of the system. We term this object the *quenched density*. By employing an imaginary-time influence functional [20–22] and its Hubbard-Stratonovich decoupling [23–25], we are able to rewrite the reduced equilibrium density not as a partial trace, but as a classical average over an ensemble of “quenched” imaginary-time propagators.

$$\bar{\rho}_S(\beta) = \mathbb{E}_\xi \left[\hat{\mathcal{T}} e^{-\int_0^\beta d\tau H_{\text{eff}}(\xi(\tau))} \right]. \quad (1)$$

In this picture, the bath is replaced by a stochastic field $\xi(\tau)$ [26–28], and the system evolves under a time-dependent effective Hamiltonian for each realization of the noise.

This formulation effectively sets the system in a frame where it can be expressed in terms of classical trajectories for the bath degrees of freedom. The Hubbard-Stratonovich transformation then allows us to exploit this to represent the bath information locally at the operator level. The framework thus provides a rigorous bridge between the nonlocal path-integral description (where the bath is exact but the system operator structure is opaque) and the local operator description required for thermodynamics.

We apply this framework to establishing a rigorous benchmark: the linearly coupled Gaussian bath in the commuting (pure-dephasing) sector [29, 30]. In this limit,

relevant for quantum nondemolition measurements and qubit dephasing [31, 32], the technical obstruction of imaginary-time time-ordering vanishes, and the quenched density's average can be performed analytically. We use this to recover the known result that the bath induces a static potential renormalization $-\lambda f^2$ (the reorganization energy) [33]. While this specific result is standard, its derivation within the quenched density framework serves two critical purposes. First, it demonstrates how the framework naturally recovers thermodynamic identities from path-integral primitives. Second, and most importantly, it provides a foundation for the generalisation of the framework to both non-commuting couplings and anharmonic environments.

The paper is structured as follows. In Sec. II we define the total Hamiltonian and reduced equilibrium objects. Sec. III develops the core of the framework: the Quenched Density representation of the imaginary-time propagator. Sec. IV connects this to the influence functional and derives the exact commuting Gaussian benchmark. Sec. V presents the numerical validation, confirming that the Quenched Density sampling, analytic prediction, and full finite-bath trace-out agree to machine precision. Sec. VI discusses how this Gaussian framework serves as the foundation for extending the theory to non-Gaussian and non-commuting environments.

II. THE TOTAL HAMILTONIAN AND THE HAMILTONIAN OF MEAN FORCE

The Hamiltonian of mean force is defined by tracing the global equilibrium operator, and therefore presupposes a concrete model for the composite Hamiltonian. We consider a system with Hamiltonian H_Q (system coordinates q), coupled to a bath with Hamiltonian H_X (bath coordinates x) through an interaction H_I , so that

$$H_{\text{tot}} = H_Q + H_X + H_I. \quad (2)$$

To obtain an explicit influence-functional representation we specialise to a Gaussian environment with linear coupling, i.e. a Caldeira-Leggett bath [21]. This choice is appropriate when the bath is large and near equilibrium, so that its fluctuations are well captured by a quadratic Hamiltonian and the leading (system-local) term in the interaction dominates.

The bath is modeled as a collection of harmonic oscillators,

$$H_X = \sum_k \left(\frac{p_k^2}{2m_k} + \frac{1}{2} m_k \omega_k^2 x_k^2 \right), \quad (3)$$

and we take a linear coupling to a system operator $f \equiv f(q)$,

$$H_I = \sum_k c_k f(q) x_k, \quad (4)$$

so that

$$H_{\text{tot}} = H_Q + \sum_k \left(\frac{p_k^2}{2m_k} + \frac{1}{2} m_k \omega_k^2 x_k^2 + c_k f(q) x_k \right). \quad (5)$$

As is standard in Caldeira-Leggett constructions, one may include a counter-term proportional to $f(q)^2$ to enforce translational invariance of the bath coordinates and to compensate the static potential renormalisation induced by the coupling [21]. In what follows we absorb any such counterterm into the definition of H_Q .

Although the microscopic bath is parametrised by $\{m_k, \omega_k, c_k\}$, its influence on the system enters only through the spectral density [30, 34],

$$J(\omega) = \frac{\pi}{2} \sum_k \frac{c_k^2}{m_k \omega_k} \delta(\omega - \omega_k), \quad (6)$$

which compactly encodes the bath two-point structure and fixes the imaginary-time influence kernel used below [20].

Having specified the composite dynamics, we define the Hamiltonian of mean force by tracing the global Hamiltonian of mean force $H_{\text{MF}}(\beta)$ as the operator whose Gibbs form reproduces the reduced equilibrium object up to a chosen normalisation. If the total system is at inverse temperature β , the unnormalised global equilibrium operator is $e^{-\beta H_{\text{tot}}}$, then the reduced (unnormalised) equilibrium operator is

$$\bar{\rho}_Q(\beta) \equiv \text{Tr}_X e^{-\beta H_{\text{tot}}}, \quad (7)$$

where Tr_X traces out the bath degrees of freedom. The Hamiltonian of mean force is defined by

$$e^{-\beta H_{\text{MF}}(\beta)} \equiv \frac{\text{Tr}_X e^{-\beta H_{\text{tot}}}}{Z_X(\beta)}, \quad Z_X(\beta) \equiv \text{Tr}_X e^{-\beta H_X}, \quad (8)$$

or equivalently

$$H_{\text{MF}}(\beta) = -(1/\beta) \log[\text{Tr}_X e^{-\beta H_{\text{tot}}}] \quad (9)$$

up to an additive scalar fixed by the choice of Z_X .

For a harmonic bath linearly coupled through a collective coordinate, the bath can be eliminated exactly, yielding the Feynman-Vernon influence functional [1, 2, 20, 22]. In equilibrium this produces a Euclidean (imaginary-time) influence functional which is bilocal in τ and whose kernel is fixed by the bath spectral density $J(\omega)$ [15, 27, 35, 36]. While this gives an exact representation of the reduced equilibrium operator, it does not generically yield a compact local expression for H_{MF} . The reduced equilibrium object is a path integral over the system degrees of freedom, rather than a simple exponential $e^{-\beta H_{\text{eff}}}$ of a time-independent operator. Extracting H_{MF} - the logarithm of this ordered object - is therefore an inverse representability problem, not merely a question of coupling strength or temperature [9, 37].

There are two logically distinct aspects to this problem. First, the bath produces an *imaginary-time nonlocal* self-interaction with a memory kernel determined by $J(\omega)$. This can be resolved with a local representation of the path integral, which can be achieved via a Hubbard-Stratonovich (HS) transformation [23, 24]. This mapping allows for a return to the operator picture by replacing the nonlocal self-interaction with a local coupling to a stochastic auxiliary field. However, this transformation introduces stochasticity that must be averaged over, and it is in this averaging process that we encounter the second obstruction: the algebraic properties of the system and its coupling. While the HS transformation allows one to return to the operator picture, to obtain the exact state now requires an averaging procedure over the stochastic auxiliary field. When $[H_Q, f] \neq 0$, evaluating this average analytically presents an (apparently) [38] insuperable difficulty. Nevertheless, in the commuting sector $[H_Q, f] = 0$, the algebraic obstruction disappears, and the averaging can be performed. Under these conditions, influence functional contains exactly the ingredients required to construct $H_{\text{MF}}(\beta)$ in closed form.

Summarising, our strategy to find $H_{\text{MF}}(\beta)$ is to rewrite the reduced state such that the contributions from the bath statistics (entering only through $J(\omega)$) and the system-coupling algebra may be cleanly separated. To achieve this, it is necessary to first develop a formal treatment connecting the quenched density representation of the reduced system to a path integral representation of the influence functional.

III. THE QUENCHED DENSITY REPRESENTATION

While many representations of reduced equilibrium exist [22, 30, 34], tracing out environmental degrees of freedom is most naturally achieved in a path integral formulation. Indeed, one of the principal advantages of this approach is that it requires no assumptions on the spectral character of the bath or the strength of its coupling to the system [34]. In this framework, tracing out the bath degrees of freedom leads to the following representation of the reduced density matrix $\bar{\rho}_Q(\beta)$ [39, 40]:

$$\langle q_f | \bar{\rho}_Q(\beta) | q_i \rangle = \frac{1}{Z_{\text{tot}}} \int \mathcal{D}q \mathcal{D}\mathbf{x} e^{-S_E[q, \mathbf{x}]}. \quad (10)$$

This is an imaginary-time path integral [39, 40], with a Euclidean action $S_E[q, \mathbf{x}]$. This contains both the bare system action, and the *influence functional* $\Gamma_E[q, \mathbf{x}]$ that remains after tracing the bath degrees of freedom. While this representation makes exact treatment of the bath possible, it comes at a conceptual cost: the path integral trades operator structure for functional structure. In particular, questions about the *operator algebra* underlying the reduced equilibrium state are obscured once

the dynamics is expressed solely in terms of paths and kernels.

If one wishes to understand when a local or otherwise restricted Hamiltonian of mean force can exist, it is therefore useful to reconnect the path-integral description to an operator-based notion of imaginary-time propagation. The key observation is that the reduced equilibrium operator may be regarded as an imaginary-time evolution over an interval of length β , but with a generator that need not be time independent. Allowing for explicit τ -dependence in the generator provides a natural bridge between the nonlocal Euclidean influence functional and a time-ordered exponential acting on the system Hilbert space.

Motivated by this, we introduce a class of imaginary-time propagators generated by a τ -dependent Hamiltonian $\bar{H}(\tau)$ acting on the system Hilbert space. With this, we define the *quenched* imaginary-time propagator $U(\beta)$ and the *quenched density* $\rho_{\bar{H}}(\beta)$ as its normalised counterpart:

$$\begin{aligned} U(\beta) &:= \hat{\tau} \exp \left[- \int_0^\beta d\tau \bar{H}(\tau) \right], \\ \rho_{\bar{H}}(\beta) &:= \frac{1}{Z_{\bar{H}}} U(\beta), \\ Z_{\bar{H}} &:= \text{Tr } U(\beta), \end{aligned} \quad (11)$$

where $\hat{\tau}$ orders increasing imaginary times (inverse temperatures) to the left [39, 40]. We refer to this construction as *quenched* as the τ -dependence is treated as fixed during the construction of the propagator - we may consider it as a quasistatic quench, where the Hamiltonian is modulated as temperature is lowered.

Having defined this object, we must make contact with the path integral representation. To this end, we assume the standard kinematic structure

$$\bar{H}(\tau) = \bar{T} + \bar{V}(\tau), \quad \bar{T} = \frac{\hat{p}^2}{2m}, \quad \bar{V}(\tau) \equiv \bar{V}(\hat{q}, \tau), \quad (12)$$

i.e. the τ -dependence enters through a potential term that is multiplicative in the $|q\rangle$ basis. (This is the case relevant to the influence-functional construction in Sec. IV.)

We now discretise $[0, \beta]$ into N slices of width $\Delta = \beta/N$ and define $\tau_j = j\Delta$, $\bar{H}_j := \bar{H}(\tau_j)$. The τ -ordered exponential is then given by the Trotter product [40, 41]

$$\hat{\tau} \exp \left[- \int_0^\beta d\tau \bar{H}(\tau) \right] = \lim_{N \rightarrow \infty} e^{-\Delta \bar{H}_N} \dots e^{-\Delta \bar{H}_1} e^{-\Delta \bar{H}_0}. \quad (13)$$

Inserting N resolutions of the identity [40] in the $|q\rangle$ basis between factors, with $q_0 = q_i$ and $q_{N+1} = q_f$, we obtain the kernel

$$Z_{\bar{H}} \bar{\rho}(q_f, q_i; \beta) = \lim_{N \rightarrow \infty} \int \prod_{j=1}^N dq_j \prod_{j=0}^N \langle q_{j+1} | e^{-\Delta \bar{H}_j} | q_j \rangle. \quad (14)$$

For small Δ we use a first-order Trotter splitting (sufficient in the continuum limit) [40]

$$e^{-\Delta(\bar{T}+\bar{V}_j)} = e^{-\Delta\bar{T}} e^{-\Delta\bar{V}_j} + \mathcal{O}(\Delta^2), \quad \bar{V}_j := \bar{V}(\hat{q}, \tau_j), \quad (15)$$

so that

$$\langle q_{j+1} | e^{-\Delta\bar{H}_j} | q_j \rangle = \langle q_{j+1} | e^{-\Delta\bar{T}} | q_j \rangle e^{-\Delta\bar{V}(q_j, \tau_j)} + \mathcal{O}(\Delta^2). \quad (16)$$

Substituting Eq. (17) and Eq. (16) into Eq. (14) yields

$$\bar{\rho}(q_f, q_i; \beta) = \lim_{\Delta \rightarrow 0} \frac{1}{Z_\beta} \int \prod_{j=1}^N dq_j \prod_{j=0}^N \left(\sqrt{\frac{m}{2\pi\Delta}} \right) \exp \left[- \sum_{j=0}^N \Delta \left(\frac{m}{2} \left(\frac{q_{j+1} - q_j}{\Delta} \right)^2 + \bar{V}(q_j, \tau_j) \right) \right]. \quad (18)$$

In the limit $N \rightarrow \infty$ (i.e. $\Delta \rightarrow 0$), Eq. (18) becomes the Euclidean path integral [39, 40, 42]

$$\rho_H(q_f, q_i; \beta) = \frac{1}{Z_H} \int_{q(0)=q_i}^{q(\beta)=q_f} \mathcal{D}q(\tau) \exp \left[- \int_0^\beta d\tau \left(\frac{m}{2} \dot{q}(\tau)^2 + \bar{V}(q(\tau), \tau) \right) \right]. \quad (19)$$

The Euclidean action $S[\bar{q}] = \int_0^\beta d\tau \left[\frac{m}{2} \dot{q}^2 + \bar{V}(q, \tau) \right]$ appearing in the exponent defines the system action referred to in subsequent sections. We note that for finite-dimensional systems, while the specific form of the path integral measure changes (e.g., to a sum over discrete trajectories or coherent states), the quenched density representation retains precisely the same form [34, 43]. The separation into a bare system action and a bath-induced stochastic driving is a consequence of the bath properties alone, and does not depend on the dimensionality of the system Hilbert space [44].

The kinetic matrix element is evaluated by inserting a momentum resolution of the identity [39, 41]:

$$\begin{aligned} \langle q_{j+1} | e^{-\Delta\bar{T}} | q_j \rangle &= \int \frac{dp}{2\pi} \langle q_{j+1} | p \rangle \langle p | q_j \rangle e^{-\Delta p^2/(2m)} \\ &= \sqrt{\frac{m}{2\pi\Delta}} \exp \left[-\frac{m}{2\Delta} (q_{j+1} - q_j)^2 \right]. \end{aligned} \quad (17)$$

IV. FROM INFLUENCE FUNCTIONALS TO THE HAMILTONIAN OF MEAN FORCE

Having established the equivalence between the quenched density and the standard Euclidean path integral in the previous section, we now outline the specific form of the reduced system when coupled to a harmonic bath. Following the influence-functional formalism [20–22] and specifically its stochastic representation for Gaussian baths [27, 35, 36], the resulting reduced density matrix admits the imaginary-time path integral representation:

$$\bar{\rho}_Q(q_f, q_i; \beta) = Z_X(\beta) \int_{q(0)=q_i}^{q(\beta)=q_f} \mathcal{D}q \exp \left[-S_Q[q] + \frac{1}{2} \int_0^\beta d\tau \int_0^\beta d\tau' f(q(\tau)) K(\tau - \tau') f(q(\tau')) \right]. \quad (20)$$

Here $Z_X(\beta) = \text{Tr}_X e^{-\beta H_X}$ is the free bath partition function, S_Q is the Euclidean system action defined and $K(\tau)$ is the imaginary-time influence kernel. In terms of the bath spectral density $J(\omega)$, the kernel is given by [20, 34]

$$K(\tau) = \frac{1}{\pi} \int_0^\infty d\omega J(\omega) \frac{\cosh \left[\omega \left(\frac{\beta}{2} - |\tau| \right) \right]}{\sinh \left(\frac{\beta\omega}{2} \right)}. \quad (21)$$

While the representation (20) is exact, it is nonlocal (bilocal) in imaginary time. This nonlocality can be resolved by a Hubbard - Stratonovich (HS) transformation [23, 24], which decouples the bilocal interaction by introducing an auxiliary Gaussian field $\xi(\tau)$. This step pro-

vides the bridge to an operator-level representation of $\bar{\rho}_Q$ as a Gaussian average over the local quenched canonical densities discussed in Sec. III:

$$\bar{\rho}_Q(\beta) = Z_X(\beta) \mathbb{E}_\xi \left[\hat{\mathcal{T}} \exp \left(- \int_0^\beta d\tau \bar{H}(\tau; \xi) \right) \right], \quad (22)$$

The statistical properties of the auxiliary field $\xi(\tau)$ are set by the influence kernel [27, 35, 36]

$$\mathbb{E}_\xi[\xi(\tau)] = 0, \quad \mathbb{E}_\xi[\xi(\tau)\xi(\tau')] = K(\tau - \tau'). \quad (23)$$

This operator representation maps directly to the quenched density framework. The challenge now lies in performing the average over ξ . To do so, we must isolate the stochastic term. Moving to the interaction picture with respect to H_Q , the propagator factorises as

$$\hat{\mathcal{T}} e^{-\int_0^\beta d\tau [H_Q - \xi(\tau)f]} = e^{-\beta H_Q} \hat{\mathcal{T}} e^{\int_0^\beta d\tau \xi(\tau)f(\tau)}, \quad (24)$$

where $f(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$. While we have successfully isolated the stochastic field $\xi(\tau)$, the coupling operator $f(\tau)$ has acquired a nontrivial imaginary-time dependence. This greatly complicates the averaging out of the stochastic term, and we defer its explicit evaluation to future work.

Here we focus instead on the commuting (QND) sector $[H_Q, f] = 0$, where the algebraic obstruction disappears: $f(\tau) = f$ is τ -independent, the $\hat{\mathcal{T}}$ ordering becomes redundant and the expression factorises:

$$\hat{\mathcal{T}} \exp \left[- \int_0^\beta d\tau \bar{H}(\tau; \xi) \right] = e^{-\beta H_Q} \exp[f X], \quad (25)$$

where $X := \int_0^\beta d\tau \xi(\tau)$ is a Gaussian variable with mean zero and variance

$$C(\beta) = \mathbb{E}_\xi[X^2] = \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau'). \quad (26)$$

Substituting the kernel Eq. (21) and performing the double imaginary-time integral gives

$$C(\beta) = \frac{2\beta}{\pi} \int_0^\infty d\omega \frac{J(\omega)}{\omega} \equiv 2\beta \lambda, \quad (27)$$

$$\lambda := \frac{1}{\pi} \int_0^\infty d\omega \frac{J(\omega)}{\omega},$$

where λ is the reorganisation energy [34, 44]. Crucially, $C(\beta)$ is *linear* in β , so the ratio $C(\beta)/(2\beta) = \lambda$ is temperature independent.

With this the Gaussian average $\mathbb{E}_\xi[e^{fX}] = \exp[\frac{1}{2}C(\beta)f^2]$ can be performed exactly, and using Eq. (22), we obtain:

$$\bar{\rho}_Q(\beta) = Z_X(\beta) e^{-\beta H_Q} \exp[\beta \lambda f^2]. \quad (28)$$

This is (up to normalisation) the canonical exponential form for a thermal state, and the Hamiltonian of mean force is identically

$$H_{\text{MF}}(\beta) = H_Q - \lambda f^2 \quad (29)$$

up to some temperature dependent constant that is fixed by the normalisation $\text{Tr} e^{-\beta H_{\text{MF}}} = 1$. The only nontrivial operator content is then the correction $-\lambda f^2$, which is temperature independent.

Interestingly λ coincides with the static potential renormalisation $\frac{1}{2} \sum_k c_k^2 / (m_k \omega_k^2)$ of the Caldeira-Leggett

model [45]. If the standard counterterm $+\lambda f^2$ has been absorbed into H_Q (as stated in Sec. II), the bath-induced shift $-\lambda f^2$ cancels it exactly. In this case, H_{MF} reduces to the bare system Hamiltonian H_Q^{bare} , and the normalised mean-force Gibbs state coincides with the bare-system Gibbs state. In other words, in the commuting sector with translational invariance enforced, the bath has no operator-level effect on the reduced equilibrium state. Conversely, if no counterterm is included, $-\lambda f^2$ represents a genuine bath-induced static renormalisation of the system potential.

Having established the quenched density representation exactly captures the reorganisation-energy result [33], we now apply it to stochastic numerical evaluation of the equilibrium state.

V. NUMERICAL EVALUATION

We now verify the predicted H_{MF} can be obtained numerically through stochastic averaging. To do so, we consider a toy five-level system $H_S = \sum_{n=0}^4 E_n |n\rangle\langle n|$ coupled to the bath via the projector $f = |4\rangle\langle 4|$. This coupling is manifestly commuting ($[H_S, f] = 0$), ensuring that the exact HMF should be

$$H_{\text{MF}} = H_S - \lambda |4\rangle\langle 4| + \text{const.} \quad (30)$$

The precise level spacings are selected randomly and kept fixed across all (β, g) simulations. For the bath, we set use $J_g(\omega) = 2g^2 \omega e^{-\omega/\omega_c}$, incorporating system-bath coupling strength directly into the parameter g . We choose $\omega_c = 1$ for all simulations. As a point of comparison, we further implement an explicit bath of K harmonic modes a_k , coupling to the system through $\sum_k a_k + a_k^\dagger$. The bath spectrum is then mimicked by discretising the interval $\omega \in [0, \omega_c]$ into mode bins with centres ω_k and widths $\Delta\omega_k$, setting $c_k^2 = \frac{2}{\pi} J_g(\omega_k) \omega_k \Delta\omega_k$. This ensures that the discretised reorganisation scale $\lambda_{\text{disc}} = \sum_k \frac{c_k^2}{2\omega_k^2}$ tracks the continuum expression $\lambda_{\text{cont}} = \frac{1}{\pi} \int_0^\infty \frac{J_g(\omega)}{\omega} d\omega$ [34]. This provides an exact reference for the reduced state $\rho_Q^{\text{ED}} = \text{Tr}_B[e^{-\beta H_{\text{tot}}}] / \text{Tr}[e^{-\beta H_{\text{tot}}}]$.

Finally, we implement the stochastic formulation of the quenched density (Eq. 22). The Gaussian noise $\xi(\tau)$ is generated by convolution of white noise with the covariance matrix of the discretised kernel $K(\tau)$, a standard technique for sampling coloured noise (see e.g. [35]). In the present model, the variance of the integrated field $X = \int_0^\beta d\tau \xi(\tau)$ is predicted to be $2\beta\lambda_{\text{disc}}$. Fig. 1 confirms this scaling, plotting the sample variance extracted from the stochastic simulations against β (panel a) and g^2 (panel b).

With this setup the core predictions of the HMF framework can be verified. Fig. 2 confirms the thermodynamic consistency of the Quenched Density framework, showing that the effective Hamiltonian correctly captures the

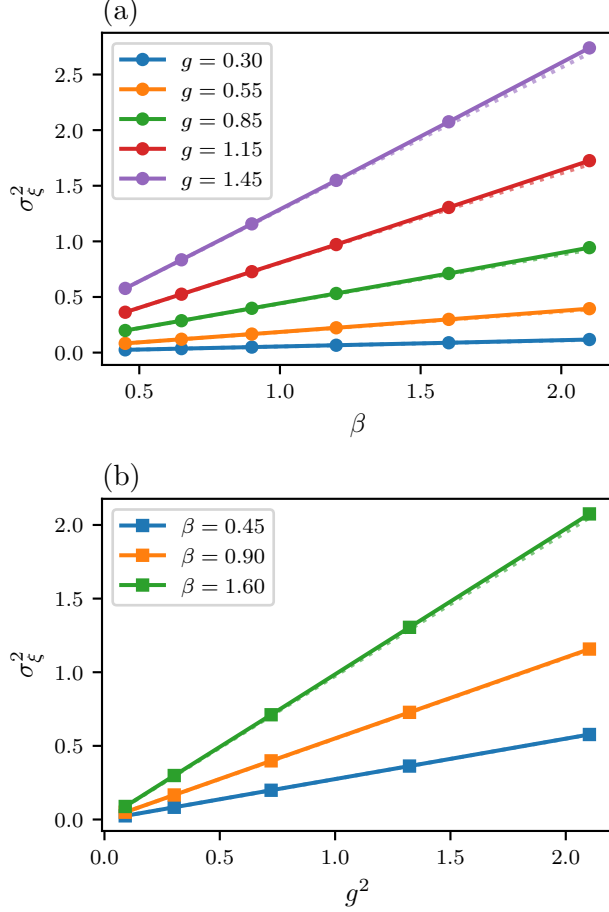


FIG. 1. Variance scaling. The sample variance σ_ξ^2 of the stochastic field grows linearly with inverse temperature β (a) and linearly with the reorganisation energy $\lambda \propto g^2$ (b). Symbols are simulation results; dotted lines are the theoretical prediction $2\beta\lambda_{\text{disc}}$.

temperature dependence of the open system, with the extracted reorganisation energy matching the theoretical value.

Fig. 3 demonstrates robustness in the strong-coupling regime, where the interaction strength rivals the system energy scales. To verify the operator structure of the HMF, we fit the numerically extracted H_{MF} to the ansatz $c_0\mathbb{I} + a_{H_S}H_S + a_PP$. As shown in panel (b), the coefficient a_P tracks the theoretical prediction $-\lambda$ perfectly, while $a_{H_S} \approx 1$ confirms that the bare system Hamiltonian is not renormalized.

Finally, Fig. 4 provides the microscopic resolution: the HMF modifies only the coupled subspace, leaving the orthogonal manifold untouched, as predicted by the operator structure of Eq. 29.

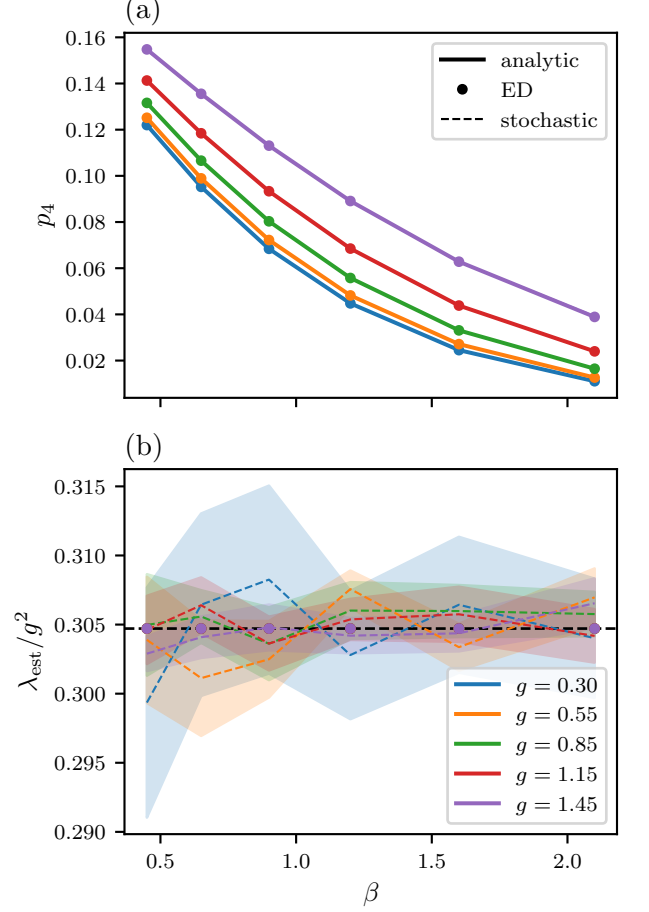


FIG. 2. Temperature dependence and scaling. (a) Population of the coupled state ($|4\rangle$) vs β for various couplings g . Circles (ED) match solid lines (analytic). (b) The extracted reorganisation energy λ_{est} collapsed by g^2 vs β . All curves hover near the theoretical thermodynamic limit (dashed lines), verifying that the bath's effect reduces to a single reorganisation energy λ .

VI. DISCUSSION AND OUTLOOK

We have presented the quenched density framework as a rigorous bridge between path-integral influence functionals and operator-level thermodynamics. By mapping the bath trace to a stochastic average over quenched propagators, we isolate the algebraic structure of the system's response from the statistical properties of the environment.

In the case of harmonic baths where H_Q and f commute, the algebraic and statistical contributions to the HMF can be cleanly partitioned: the bath statistics collapse to a scalar reorganisation energy λ , and the system algebra reduces to a static potential shift $-\lambda f^2$. Our numerical benchmarks (Sec. V) confirm that this framework reproduces the exact finite-bath physics, regardless

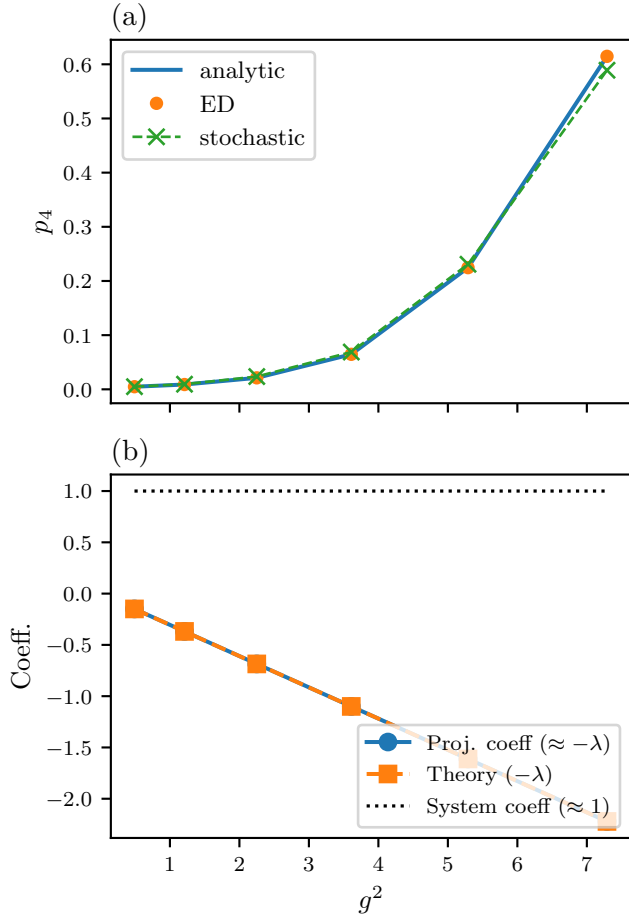


FIG. 3. Strong-coupling benchmark. (a) Population p_4 vs g^2 at fixed β , showing excellent agreement over the full range. (b) Coefficients of the HMF in the system basis vs g^2 . The projector component $P = |4\rangle\langle 4|$ scales as $-\lambda$ (orange/blue), while H_S remains unrenormalised (dotted), confirming $H_{MF} = H_S - \lambda f^2$.

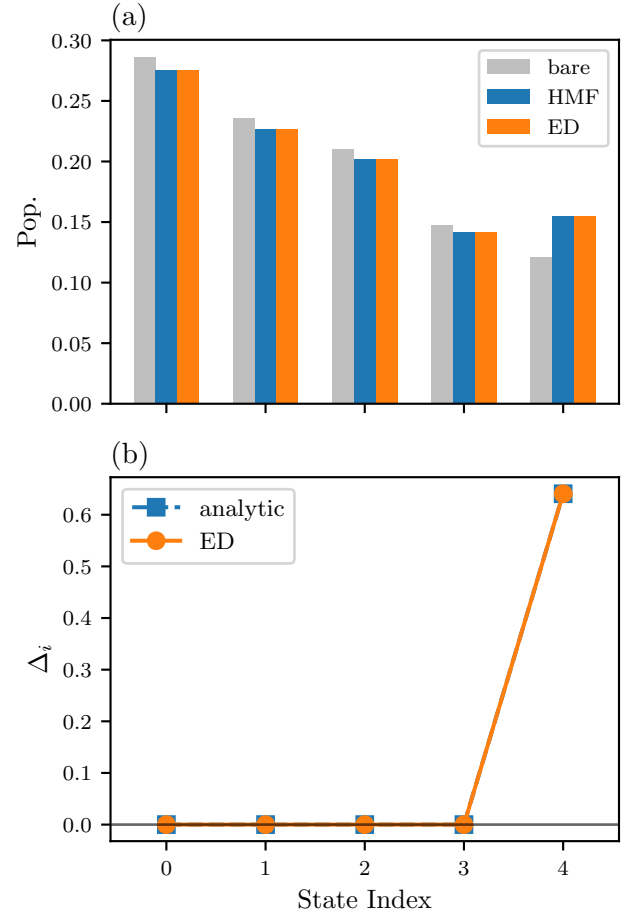


FIG. 4. Microscopic validation. (a) Detailed population snapshot at fixed β, g : "Analytic HMF" (blue) corrects the "Bare Gibbs" (grey) to match ED (orange). (b) Effective energy shifts Δ_i for each level; only the coupled level ($|4\rangle$) shifts, by exactly $-\lambda$. Non-coupled levels remain unperturbed.

of coupling strength.

The exact result derived here is hardly novel, and in isolation can hardly justify the formal machinery used to derive it. In this regard, our treatment is incomplete. The power of the quenched density lies in its modularity, which naturally accommodates both more general environments, and system bath-couplings (where $[H_S, f] \neq 0$). In the latter case, the challenge shifts from the statistical generation of noise to the Lie-algebraic properties of the time-ordered effective propagator. The framework thus clearly delineates the statistical complexity of the bath from the dynamical complexity of the system response. In forthcoming work, we shall address these generalisations. Even in this simplified case however, we see the shape of the answer to the question with which we began. What governs equilibrium? Algebra.

-
- [1] P. Hänggi, G.-L. Ingold, and P. Talkner, Finite quantum dissipation: The challenge of obtaining specific heat, *New Journal of Physics* **10**, 115008 (2008).
 - [2] G.-L. Ingold, P. Hänggi, and P. Talkner, Specific heat anomalies of open quantum systems, *Physical Review E* **79**, 061105 (2009).
 - [3] M. Campisi, P. Talkner, and P. Hänggi, Fluctuation Theorem for Arbitrary Open Quantum Systems, *Physical Review Letters* **102**, 210401 (2009).
 - [4] C. Jarzynski, Nonequilibrium work theorem for a system strongly coupled to a thermal environment, *Journal of Statistical Mechanics: Theory and Experiment* **2004**, P09005 (2004).
 - [5] M. Esposito, M. A. Ochoa, and M. Galperin, Nature of heat in strongly coupled open quantum systems, *Physical Review B* **92**, 235440 (2015).
 - [6] Á. Rivas, Strong Coupling Thermodynamics of Open Quantum Systems, *Physical Review Letters* **124**, 160601 (2020).

- (2020).
- [7] P. Strasberg and M. Esposito, Measurability of nonequilibrium thermodynamics in terms of the Hamiltonian of mean force, *Physical Review E* **101**, 050101 (2020).
 - [8] J. Anders, C. Sait, and S. Horsley, Versatile three-dimensional quantum spin dynamics equation with guaranteed fluctuation-dissipation link, *Phys. Rev. Research* **4**, 023037 (2022).
 - [9] P. Talkner and P. Hänggi, *Colloquium* : Statistical mechanics and thermodynamics at strong coupling: Quantum and classical, *Reviews of Modern Physics* **92**, 041002 (2020).
 - [10] A. S. Trushechkin, M. Merkli, J. D. Cresser, and J. Anders, Open quantum system dynamics and the mean force Gibbs state, *AVS Quantum Science* **4**, 012301 (2022).
 - [11] U. Seifert, First and Second Law of Thermodynamics at Strong Coupling, *Physical Review Letters* **116**, 020601 (2016).
 - [12] J. D. Cresser and J. Anders, Weak and Ultrastrong Coupling Limits of the Quantum Mean Force Gibbs State, *Physical Review Letters* **127**, 250601 (2021).
 - [13] X. Du, J. Wang, and Y. Ma, Generalized Hamiltonian of mean-force approach to open quantum systems coupled to finite baths in thermoequilibrium, *Physical Review E* **111**, 024139 (2025).
 - [14] A. Nazir and G. Schaller, The reaction coordinate mapping in quantum thermodynamics, *Thermodynamics in the Quantum Regime* , 551 (2018).
 - [15] J. M. Moix, Y. Zhao, and J. Cao, Equilibrium-reduced density matrix formulation: Influence of noise, disorder, and temperature on localization in excitonic systems, *Physical Review B* **85**, 115412 (2012).
 - [16] X. Chen, The rigorous stochastic matrix multiplication scheme for the calculations of reduced equilibrium density matrices of open multilevel quantum systems, *The Journal of Chemical Physics* **140**, 154101 (2014).
 - [17] N. Makri, Exploiting classical decoherence in dissipative quantum dynamics: Memory, phonon emission, and the blip sum, *Chemical Physics Letters* **593**, 93 (2014).
 - [18] Y. Tanimura, Reduced hierarchical equations of motion in real and imaginary time: Correlated initial states and thermodynamic quantities, *The Journal of Chemical Physics* **141**, 044114 (2014).
 - [19] L. Song and Q. Shi, Calculation of correlated initial state in the hierarchical equations of motion method using an imaginary time path integral approach, *The Journal of Chemical Physics* **143**, 194106 (2015).
 - [20] R. P. Feynman and F. L. Vernon, The theory of a general quantum system interacting with a linear dissipative system, *Annals of Physics* **24**, 118 (1963).
 - [21] A. O. Caldeira and A. J. Leggett, Quantum tunnelling in a dissipative system, *Annals of Physics* **149**, 374 (1983).
 - [22] H. Grabert, P. Schramm, and G.-L. Ingold, Quantum Brownian motion: The functional integral approach, *Physics Reports* **168**, 115 (1988).
 - [23] J. Hubbard, Calculation of Partition Functions, *Physical Review Letters* **3**, 77 (1959).
 - [24] R. L. Stratonovich, On a method of calculating quantum distribution functions, *Soviet Physics Doklady* **2**, 416 (1957).
 - [25] J. T. Stockburger and H. Grabert, Exact c -Number Representation of Non-Markovian Quantum Dissipation, *Physical Review Letters* **88**, 170407 (2002).
 - [26] J. T. Stockburger, Simulating spin-boson dynamics with stochastic Liouville–von Neumann equations, *Chemical Physics* **296**, 159 (2004).
 - [27] G. M. G. McCaul, C. D. Lorenz, and L. Kantorovich, Partition-free approach to open quantum systems in harmonic environments: An exact stochastic Liouville equation, *Physical Review B* **95**, 125124 (2017).
 - [28] M. Wiedmann, J. T. Stockburger, and J. Ankerhold, Non-Markovian dynamics of a quantum heat engine: Out-of-equilibrium operation and thermal coupling control, *New Journal of Physics* **22**, 033007 (2020).
 - [29] A. A. Clerk, M. H. Devoret, S. M. Girvin, F. Marquardt, and R. J. Schoelkopf, Introduction to quantum noise, measurement, and amplification, *Rev. Mod. Phys.* **82**, 1155 (2010).
 - [30] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, Oxford, 2002).
 - [31] Y. Makhlin, G. Schön, and A. Shnirman, Quantum-state engineering with josephson-junction devices, *Rev. Mod. Phys.* **73**, 357 (2001).
 - [32] G. M. Palma, K.-A. Suominen, and A. K. Ekert, Quantum computers and dissipation, *Proc. Roy. Soc. Lond. A* **452**, 567 (1996).
 - [33] M. Campisi, P. Talkner, and P. Hänggi, Thermodynamics and fluctuation theorems for a strongly coupled open quantum system: An exactly solvable case, *Journal of Physics A: Mathematical and Theoretical* **42**, 392002 (2009).
 - [34] U. Weiss, *Quantum Dissipative Systems*, 4th ed. (World Scientific, Singapore, 2012).
 - [35] G. M. G. McCaul, C. D. Lorenz, and L. Kantorovich, Driving spin-boson models from equilibrium using exact quantum dynamics, *Physical Review B* **97**, 224310 (2018).
 - [36] G. McCaul and Denys. I. Bondar, How to win friends and influence functionals: Deducing stochasticity from deterministic dynamics, *The European Physical Journal Special Topics* **230**, 733 (2021).
 - [37] K. V. Hovhannisyanyan, S. Nemati, C. Henkel, and J. Anders, Hamiltonian of mean force for strongly-coupled systems, *Physical Review E* **102**, 012104 (2020).
 - [38] In the sequel we shall revisit and resolve this issue.
 - [39] R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals* (McGraw-Hill, New York, 1965).
 - [40] H. Kleinert, *Path Integrals in Quantum Mechanics, Statistics, Polymer Physics, and Financial Markets*, 5th ed. (World Scientific, Singapore, 2009).
 - [41] L. S. Schulman, *Techniques and Applications of Path Integration* (Wiley, New York, 1981).
 - [42] C. Grosche and F. Steiner, *Handbook of Feynman Path Integrals*, Springer Tracts in Modern Physics, Vol. 145 (Springer, Berlin, 1998).
 - [43] A. Altland and B. D. Simons, *Condensed Matter Field Theory*, 2nd ed. (Cambridge University Press, 2010).
 - [44] A. J. Leggett, S. Chakravarty, A. T. Dorsey, M. P. A. Fisher, A. Garg, and W. Zwerger, Dynamics of the dissipative two-state system, *Reviews of Modern Physics* **59**, 1 (1987).
 - [45] L. A. Correa and J. Glatthard, Potential renormalisation, Lamb shift and mean-force Gibbs state – to shift or not to shift?, *Quantum* **9**, 1868 (2025).